

Workshop on Advanced Sampling Methodologies

Welcome and Introduction

Dr. Gianluca Boo, WorldPop, University of Southampton

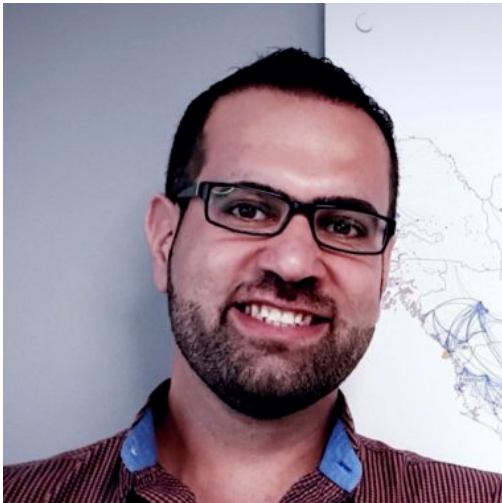
2025-09-17

Welcome!

Team

Dr. Sarchil Quader

- Senior Enterprise Fellow at WorldPop, University of Southampton.
- Develop research to improve population mapping and agricultural monitoring in low- and middle-income countries using big data, such as remote sensing.



Dr. Gianluca Boo

- Senior Research Fellow at WorldPop, University of Southampton.
- Develop research to improve population mapping in low- and middle-income countries using statistical modelling, particularly Bayesian hierarchical models.



Workshop

Objectives

- Strengthen **geospatial and sampling** capacity of Pakistan Bureau of Statistics (PBS).
- Introduce AI-assisted sampling to **improve efficiency, representativeness, and inclusivity**.
- Align **methods with Government priorities** and international standards for comparability.
- Enhance **policy relevance of PDHS data** and enable cross-country analysis.

Workshop

Topics

- Foundations of **survey sampling** theory and practice.
- **Geospatial data** processing and updating census/sampling frames.
- Use of **Earth Observation (EO)** data for spatial stratification.
- Application of **AI/ML methods** (clustering, prediction) in sample design.
- **Simulation and optimization** of sampling strategies using R.
- Development of R Shiny applications to **evaluate and visualize sampling designs**.
- Hands-on practice with **survey analysis software** and calibration techniques.
- Addressing **cost-effectiveness and nonresponse** adjustments in survey design.

Workshop

Expected Outcomes

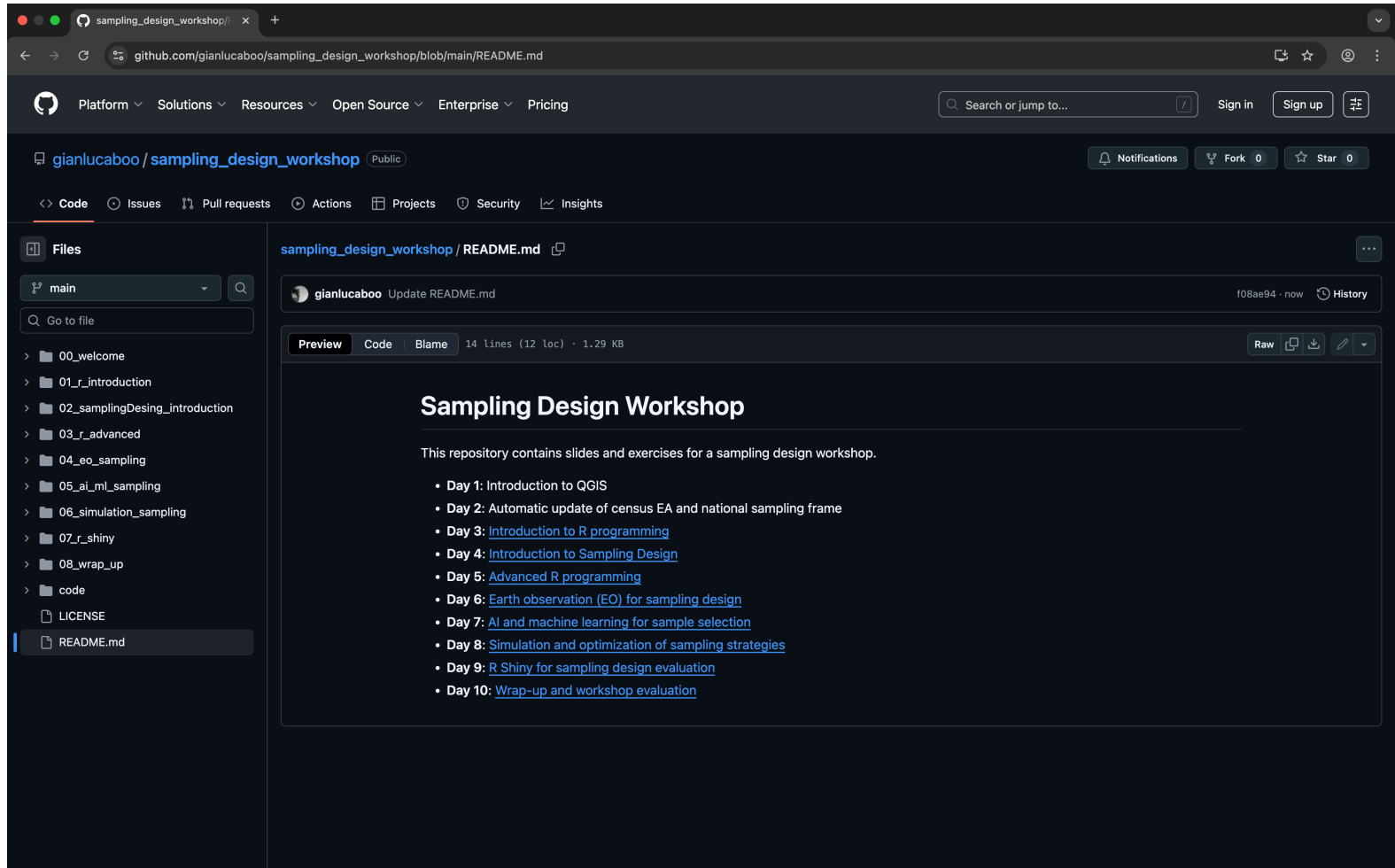
- Enhanced PBS ability to **design robust, adaptive, and cost-effective samples**.
- Practical **skills in R, GIS, EO data, and AI/ML tools**.
- Prototype **interactive tools (Shiny apps) for survey design evaluation**.
- Improved readiness to implement the upcoming PDHS with **modern, AI-driven approaches**.

Agenda

- **Day 1:** Introduction to QGIS
- **Day 2:** Automatic update of census EA and national sampling frame
- **Day 3:** Introduction to R programming
- **Day 4:** Introduction to Sampling Design
- **Day 5:** Advanced R programming
- **Day 6:** Earth observation (EO) for sampling design
- **Day 7:** AI and machine learning for sample selection
- **Day 8:** Simulation and optimization of sampling strategies
- **Day 9:** R Shiny for sampling design evaluation
- **Day 10:** Wrap-up and workshop evaluation

Course materials

All the course materials are available on a dedicated GitHub repository.



The screenshot displays the GitHub web interface for the repository `gianlucaboo/sampling_design_workshop`. The repository is public and has 0 forks and 0 stars. The `main` branch is selected. The file `README.md` is open, showing a preview of the content. The README title is **Sampling Design Workshop**. The content states: "This repository contains slides and exercises for a sampling design workshop." followed by a list of 10 days of content:

- Day 1: Introduction to QGIS
- Day 2: Automatic update of census EA and national sampling frame
- Day 3: [Introduction to R programming](#)
- Day 4: [Introduction to Sampling Design](#)
- Day 5: [Advanced R programming](#)
- Day 6: [Earth observation \(EO\) for sampling design](#)
- Day 7: [AI and machine learning for sample selection](#)
- Day 8: [Simulation and optimization of sampling strategies](#)
- Day 9: [R Shiny for sampling design evaluation](#)
- Day 10: [Wrap-up and workshop evaluation](#)

The left sidebar shows the file structure of the repository, including folders for each day (00_welcome to 08_wrap_up), a `code` folder, and files `LICENSE` and `README.md`.

Let's get started!